Example 2.81

Evaluate the integral as the limit of a sum: $\int_0^1 x\,dx$

Solution:

$$\int_a^b f(x)dx = \lim_{\substack{n\to\infty \\ h\to 0}} \sum_{r=1}^{n} h\ f(a+rh)$$

Here $a=0,\ b=1,\ h=\frac{b-a}{n}=\frac{1-0}{n}=\frac{1}{n}$ and $f(x)=x$

Now $f(a+rh)=f\left(0+\frac{r}{n}\right)=f\left(\frac{r}{n}\right)=\frac{r}{n}$

On substituting in (1) we have

$$\begin{aligned}\int_0^1 x\,dx &= \lim_{n\to\infty} \sum_{r=1}^{n} \frac{1}{n}.\frac{r}{n} \\ &= \lim_{n\to\infty} \frac{1}{n^2} \sum_{r=1}^{n} r \\ &= \lim_{n\to\infty} \frac{1}{n^2}\ \frac{n(n+1)}{2} \\ &= \lim_{n\to\infty} \frac{1}{n^2}\cdot \frac{n^2\left(1+\frac{1}{n}\right)}{2} \\ &= \frac{1+0}{2}=\frac{1}{2}\end{aligned}$$

$$\therefore \quad \int_0^1 x\,dx = \frac{1}{2}$$

Example 2.82

Evaluate the integral as the limit of a sum: $\int_1^2 (2x+1)\,dx$

Solution:

$$\int_a^b f(x)\,dx = \lim_{\substack{n\to\infty \\ h\to 0}} \sum_{r=1}^{n} h\ f(a+rh)$$

Here $a=1,\ b=2,\ h=\frac{b-a}{n}=\frac{2-1}{n}=\frac{1}{n}$

and $f(x)=2x+1$, $f(a+rh) = f\left(1+\frac{r}{n}\right)$

$$= 2\left(1+\frac{r}{n}\right)+1$$

$$= 2+\frac{2r}{n}+1$$

$$f(a+rh) = 3+\frac{2r}{n}$$

$$\begin{aligned}\int_1^2 f(x)\,dx &= \lim_{n\to\infty} \sum_{r=1}^{n} \frac{1}{n}\left(3+\frac{2r}{n}\right) \\ &= \lim_{n\to\infty} \sum_{r=1}^{n}\left(\frac{3}{n}+\frac{2r}{n^2}\right) \\ &= \lim_{n\to\infty}\left[\frac{3}{n}\sum_{r=1}^{n}1+\frac{2}{n^2}\sum_{r=1}^{n}r\right] \\ &= \lim_{n\to\infty}\left[\frac{3}{n}(n)+\frac{2}{n^2}\frac{n(n+1)}{2}\right] \\ &= 3+\lim_{n\to\infty}\left(1+\frac{1}{n}\right)\end{aligned}$$

$$\int_1^2 f(x)\,dx = 3+1=4$$

Example 2.83

Evaluate the integral as the limit of a sum: $\int_1^2 x^2\,dx$

Solution:

$$\int_a^b f(x)dx = \lim_{\substack{n\to\infty \\ h\to 0}} \sum_{r=1}^{n} h\ f(a+rh)$$

Here $a=1,\ b=2,\ h=\frac{b-a}{n}=\frac{2-1}{n}=\frac{1}{n}$ and $f(x)=x^2$

Now $f(a+rh)= f\left(1+\frac{r}{n}\right) = \left(1+\frac{r}{n}\right)^2$

$$= 1+\frac{2r}{n}+\frac{r^2}{n^2}$$

$$\therefore \quad \int_1^2 x^2\,dx = \lim_{n\to\infty} \sum_{r=1}^{n} \frac{1}{n}\left(1+\frac{2r}{n}+\frac{r^2}{n^2}\right)$$

$$= \lim_{n\to\infty} \sum_{r=1}^{n}\left(\frac{1}{n}+\frac{2r}{n^2}+\frac{r^2}{n^3}\right)$$

$$= \lim_{n\to\infty}\left(\frac{1}{n}\sum_{r=1}^{n}1+\frac{2}{n^2}\sum_{r=1}^{n}r+\frac{1}{n^3}\sum_{r=1}^{n}r^2\right)$$

$$= \lim_{n\to\infty}\left(\frac{1}{n}(n)+\frac{2}{n^2}\frac{n(n+1)}{2}+\frac{1}{n^3}\frac{n(n+1)(2n+1)}{6}\right)$$

$$= \lim_{n\to\infty}\left[1+\left(1+\frac{1}{n}\right)+\frac{\left(1+\frac{1}{n}\right)\left(2+\frac{1}{n}\right)}{6}\right]$$

XII Std - Business Maths & Stat EM Chapter 2.indd 52 2/28/2020 5:41:29 PM